A Hairpin Counterexample to the Unbounded-Interval Version of Lemma 4.2

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Status

This packet records a candidate counterexample, likely valid. It gives a negative answer to the question in arXiv:2012.12984, page 14, asking whether Lemma 4.2 remains true when the parameter interval of the curve is unbounded. The counterexample already lives in the Hilbert plane $X = \mathbb{R}^2$, hence in a dual of a separable Banach space.

Source Passage

After proving Lemma 4.2 for compactly parametrized curves, the author explains that the proof may fail on unbounded domains because one must choose a large enough interval whose image remains inside a ball, and the known input gives only local bi-Lipschitz control. The paper then asks:

> *Remark* 4.3. It is at this point that our proof of Theorem 1.1 might not carry over to the case in which the domain of γ is unbounded. (See Remark 3.11.) One must be able to choose a large enough interval in the domain which maps inside B. In particular, γ should be injective on this interval so that (4.3) can hold. However, we only know that γ is *locally* bi-Lipschitz. This leads to the following question whose answer would determine whether or not Theorem 1.1 can be proven on unbounded domains.
>
> **Question.** Does Lemma 4.2 hold when γ is defined on an unbounded interval?
>
> We now return to the proof of Theorem 1.1. Fix a ball B centered on Γ and choose $G \subset B \cap \Gamma$ as in Lemma 4.2. In order to apply Lemma 4.1 and conclude the proof,

The Question and the Relevant Condition

In the notation of the source paper, Lemma 4.2 asserts that for every ball B centered on $\Gamma = \gamma([0, 1])$ there is an interval $J = [a, b]$ such that $G = \gamma(J) \subset B \cap \Gamma$, γ is bi-Lipschitz on J, and

$$\mathcal{H}^1(G) \geq \theta\, \mathcal{H}^1(B \cap \Gamma)$$

for a constant $\theta > 0$ independent of B. The selected interval must also satisfy the source's condition (4.3):

$$\operatorname{diam}\big(J \cap \gamma^{-1}(B(y,s) \cap G)\big) \leq \frac{2s}{\inf \|\gamma'\|} \qquad (y \in G,\ s > 0).$$

For an arc-length parametrization, this implies

$$|u - v| \leq 2\|\gamma(u) - \gamma(v)\| \qquad (u, v \in J). \tag{1}$$

Indeed, apply (4.3) with $y = \gamma(u)$, let $s \downarrow \|\gamma(u) - \gamma(v)\|$, and use $\inf \|\gamma'\| = 1$.

Counterexample Statement

Theorem 1. *There is a $C^{1,1}$, arc-length parametrized curve $\gamma : [0,\infty) \to \mathbb{R}^2$ such that $\Gamma = \gamma([0,\infty))$ is 1-Ahlfors-David regular, but no constant $\theta > 0$ can make the conclusion of Lemma 4.2 true for all balls centered on Γ.*

Proof Intuition

The compact proof of Lemma 4.2 uses finite total length: once $\mathcal{H}^1(B \cap \Gamma)$ is controlled by the radius, one chooses a parameter interval of proportional length and keeps it shorter than the uniform local bi-Lipschitz scale. On an unbounded interval the ball may contain many separate visits of the curve. Ahlfors regularity controls the total amount of curve in the ball, but it does not force those visits to occur during one long bi-Lipschitz parameter interval.

The construction below is a smooth infinite ladder of hairpins. Each hairpin has two long nearly parallel strands separated by a tiny distance ε. A ball around the initial point contains the first N hairpins and therefore length comparable to N. However any interval satisfying (4.3) can cross at most one hairpin: otherwise it contains two points on opposite strands which are ε-close in the plane but order-one apart in parameter time, contradicting (1). Hence the largest admissible G has uniformly bounded length while $\mathcal{H}^1(B \cap \Gamma) \to \infty$.

Construction

Fix $h = 4$, choose $0 < \varepsilon < 1/100$, and let $\rho = \varepsilon/100$. Start from the polygonal ray P through the successive vertices

$$A_k = (0, hk), \quad B_k = (1, hk), \quad C_k = (1, hk + \varepsilon), \quad D_k = (0, hk + \varepsilon), \quad A_{k+1} = (0, h(k+1))$$

for $k = 0, 1, 2, \ldots$. Thus one block travels right along a unit horizontal strand, moves up by ε, travels left along a parallel strand, and then goes vertically up to the next block.

Replace each corner of P by the standard tangent circular rounding of radius ρ. Since $\rho \ll \varepsilon$, the rounded curve still contains the middle portions of the two horizontal strands

$$S_k^0 = \{(x, hk) : 2\rho \leq x \leq 1 - 2\rho\}, \qquad S_k^1 = \{(x, hk + \varepsilon) : 2\rho \leq x \leq 1 - 2\rho\}.$$

Let Γ be this rounded ray and let $\gamma : [0,\infty) \to \mathbb{R}^2$ be its arc-length parametrization, starting at A_0. Then γ is $C^{1,1}$, $\|\gamma'\| \equiv 1$, and γ' is Lipschitz with constant at most $1/\rho$ on each rounded arc and constant on the straight pieces.

Lemma 1. Γ *is 1-Ahlfors-David regular.*

Proof. The curve has a uniformly repeating geometry. In each vertical slab

$$Q_k = [-\rho, 1+\rho] \times [hk - \rho, h(k+1) + \rho]$$

the length of $\Gamma \cap Q_k$ is bounded above and below by absolute positive constants, and adjacent slabs overlap only a bounded number of times.

For the upper bound, first consider $0 < r \leq 10$. A ball of radius r meets only a bounded number of straight or circular pieces, and each such piece contributes at most Cr length; the constant is uniform because the rounding radius is fixed and every ball meets only boundedly many nearby parallel pieces. For $r > 10$, the ball meets only $O(r/h)$ slabs, each of uniformly bounded length, so $\mathcal{H}^1(\Gamma \cap B(x,r)) \leq Cr$.

For the lower bound, if $r \leq 10$, the point $x \in \Gamma$ lies on a straight segment or on a circular arc of radius ρ, so a subarc of length cr remains in $B(x,r)$. If $r > 10$, the strip width is bounded and the vertical connector in every block guarantees that $B(x,r)$ contains the curve in at least cr/h consecutive slabs, each contributing a fixed positive amount of length. Thus

$$C^{-1} r \leq \mathcal{H}^1(\Gamma \cap B(x,r)) \leq Cr$$

for all $x \in \Gamma$ and $r > 0$, after changing C once. □

Lemma 2. *There is a constant L_*, depending only on h, ε, ρ, such that every compact interval $J \subset [0,\infty)$ satisfying* (1) *has length at most L_*.*

Proof. Let $p_k = (1/2, hk) \in S_k^0$ and $q_k = (1/2, hk + \varepsilon) \in S_k^1$. These two points are traversed in the same block, with p_k appearing before q_k. Their Euclidean distance is ε. The arclength between their parameter times is at least

$$(1/2 - 2\rho) + \varepsilon + (1/2 - 2\rho) > 3/4,$$

where the small loss accounts for the rounded corner at the right end. Since $2\varepsilon < 1/50 < 3/4$, no interval satisfying (1) can contain both parameter times corresponding to p_k and q_k.

The blocks occur in their natural order along the parameter. If an interval meets more than a uniformly bounded number of consecutive blocks, then it contains both p_k and q_k for at least one complete intermediate block, contradicting the preceding paragraph. More explicitly, an interval avoiding all forbidden pairs (p_k, q_k) is contained between p_k and q_{k+1} for some k, up to the initial and terminal partial pieces of adjacent blocks. The length of one block is bounded by $h+3$, so one may take

$$L_* = 3(h+3).$$

□

Proof of the theorem. For $N \geq 1$, set

$$B_N = B(A_0, hN + 2).$$

The center A_0 belongs to Γ. The ball B_N contains the first N rounded hairpin blocks, and each block has length at least $h+1$. Therefore

$$\mathcal{H}^1(B_N \cap \Gamma) \geq (h+1)N.$$

Let $J \subset [0,\infty)$ be any interval for which $G = \gamma(J) \subset B_N \cap \Gamma$ and condition (4.3) of Lemma 4.2 holds. Since γ is parametrized by arclength, (4.3) implies (1); Lemma 2 gives

$$\mathcal{H}^1(G) \leq |J| \leq L_*.$$

Consequently

$$\frac{\mathcal{H}^1(G)}{\mathcal{H}^1(B_N \cap \Gamma)} \le \frac{L_*}{(h+1)N} \longrightarrow 0.$$

No positive constant θ can work simultaneously for all balls centered on Γ. This disproves the unbounded-interval version of Lemma 4.2. $\square$

Verification Notes

The argument is geometric and does not depend on numerical computation. The only code in this packet is the reproducible crop script `code/make_open_problem_crop.py`, used to extract the source question from the downloaded arXiv PDF.

Novelty and Literature Check

The run indexes were searched for arXiv:2012.12984, the paper title, and the phrases "Singular integrals", "$C^{1,\alpha}_{w^*}$", "unbounded interval", and "Lemma 4.2". No prior solution, attempt, or proof-gap packet for this source question was found.

A bounded external web search on 2026-06-23 for exact phrases including "Does Lemma 4.2 hold when gamma is defined on an unbounded interval", "l-chooseG unbounded interval", and the source title with "unbounded" surfaced the source paper but no separate later answer. Novelty confidence is moderate: the search was exact-phrase oriented rather than exhaustive, but the counterexample is elementary and addresses the mechanism highlighted by the source remark.

Limitations

This counterexample disproves Lemma 4.2 as an interval-selection statement on unbounded domains under the natural regularity hypotheses. It does not rule out other big-piece selection principles on unbounded regular curves, nor does it rule out an unbounded-domain version of the main singular-integral theorem under an additional global geometric hypothesis such as a chord-arc or bounded turning condition.

Human Review Recommendation

Reviewers should first check Lemma 2: condition (4.3) is the source's extra preimage-diameter condition, and it is what prevents an admissible interval from passing through a hairpin. Once that is accepted, the failure of a uniform θ follows immediately from the balls B_N.

References

[1] S. Zimmerman, *Singular integrals on $C^{1,\alpha}_{w^*}$ regular curves in Banach duals*, arXiv:2012.12984.